3.1.5 Implementing agency and embodiment

The remaining indicators are agency and embodiment. Reinforcement learning is arguably sufficient for agency as we have characterised it ("learning from feedback and selecting outputs so as to pursue goals"), so meeting this part of indicator AE-1 may be very straightforward. Reinforcement learning is very widely used in current AI. To recap the argument for the claim that RL is sufficient for agency, the basic task in RL is for the system to learn to maximise cumulative reward in an environment in which its outputs affect not only the immediate reward it receives but also its opportunities for future reward. RL algorithms are, therefore, designed to be sensitive to the consequences of outputs, making the system more likely to repeat outputs that lead to greater reward over multiple time-steps. This means that there is a contrast between RL and other forms of machine learning. Like systems trained by supervised or self-supervised learning, RL systems gradually come to approximate a desired input-output function. However, in RL what makes this the desired function is that the outputs are conducive to goals that can only be achieved by extended episodes of interaction between the system and the environment. Crucially, RL systems learn these functions by virtue of their sensitivity to relationships between outputs, subsequent inputs, and rewards. Model-based RL systems learn models of these relationships, but model-free systems are also sensitive to them. So RL systems learn and select actions so as to pursue goals (Butlin 2022, 2023).

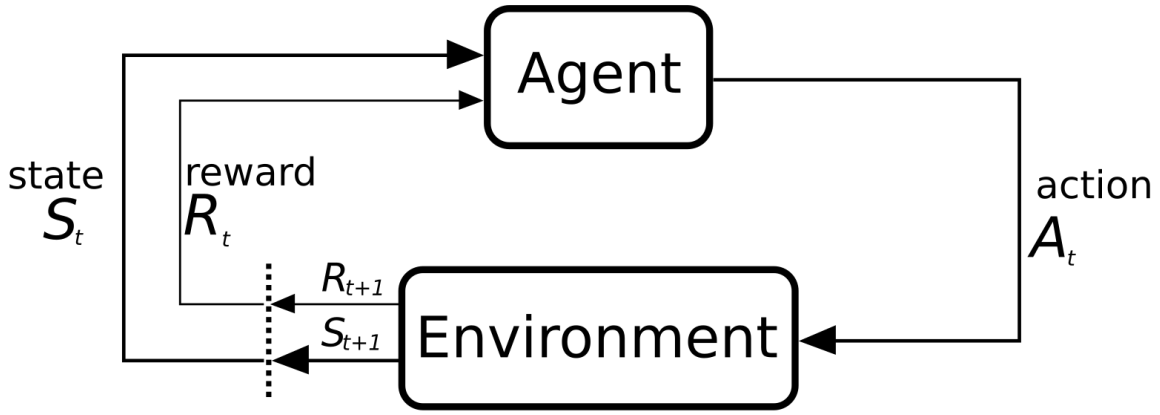

Figure 4: Reinforcement learning diagram of a Markov decision process based on a figure from *Reinforcement Learning: An Introduction* (second edition). by Sutton and Barto. 2020. Creative Commons Attribution-Share Alike 4.0 International. Reprinted with permission.

The second part of indicator AE-1 says that the probability of consciousness is raised to a greater degree if systems exhibit "flexible responsiveness to competing goals". The example motivating this clause is an animal balancing multiple homeostatic drives: this requires prioritisation which is sensitive to shifting circumstances. Several computational approaches to this problem have been explored, including in the context of reinforcement learning (Keramati & Gutkin 2014, Juechems & Summerfield 2019, Andersson et al. 2019). One proposal is to use multiple modules that learn independently about how to maximise distinct reward functions, with each of these modules assigning scores to possible actions, and then to pick the action with the highest total score (Dulberg et al. 2023). In the homeostatic case, these reward functions might each correspond to a homeostatic drive, with large penalties for allowing any one of them to depart too far from a set point.

Our embodiment indicator, AE-2, states that systems should use output-input models (also known as forward models) for perception or control. There are specific uses of such models, which neuroscientists have identified in humans, that are particularly associated with embodiment. In perception, predictions of the sensory effects of the system's own actions can be used to distinguish changes to sensory stimulation caused by these actions from those caused by events in the environment. Systems that engage in this form of inference implicitly or explicitly distinguish themselves from their environments. In motor control, forward models can be used for state estimation and feedback control, enabling rapid adjustments of complex effectors. It is important to distinguish these uses of output-input models from other uses, such as planning, which do not imply embodiment.

Learning output-input models for tasks related to perception and control is common, but there are few examples of current AI systems which meet these specific descriptions. For example, video prediction is a much-studied task (Oprea et al. 2020), and this includes predicting how visual input will develop conditional on the system's outputs (Finn et al. 2016). But this is not sufficient for the kind of use in perception that we have described, which also includes making sense of exogenous changes which are not predictable from outputs. Output-input models are used in model-based reinforcement learning for control, facilitating successful control of quadrotor drones (Becker-Ehmck et al. 2020) and other robots (Wu et al. 2022). But it is uncertain whether the models are used in these cases for control-specific purposes rather than for planning, a topic that we explore further in section 3.2.2.

One good example of recent research which does employ a forward model for a purpose specific to embodied systems is the system described in Friedrich et al. (2021). In this system, Kalman filtering (Todorov & Jordan 2002) was used to combine a forward model with sensory inputs, which were subject to a time delay, in order to estimate the current state of the system in its environment. These estimates were then used in ongoing motor control, although only of relatively simple effectors in virtual environments.

3.2 Case Studies of Current Systems

3.2.1 Case studies for GWT

In practice, it is not always immediately obvious whether a given AI system possesses one of the indicator properties. One reason for this is that we have not given absolutely precise definitions for each indicator. Another is that how deep learning systems work, including what they represent at intermediate layers, is often not transparent. In this section and section 3.2.2 we present case studies illustrating the use of the indicators to assess current AI systems. Here we focus on the GWT indicators (GWT-1 through GWT-4), and on two kinds of systems that are notable for different reasons. These systems are Transformer-based large language models (LLMs) such as GPT-3 (Brown et al. 2020), GPT-4 (OpenAI 2023), and LaMDA (Thoppilan et al. 2022), which are notable for their remarkable performance on natural language tasks and the public attention they have attracted; and Perceiver (Jaegle et al. 2021a) and Perceiver IO (Jaegle et al. 2021b), which are notable because Juliani et al. (2022) argue that they "implement a functioning global workspace". Neither of these kinds of systems was designed to implement a global workspace, but there are arguments to be made that they each possess some of the GWT indicator properties.

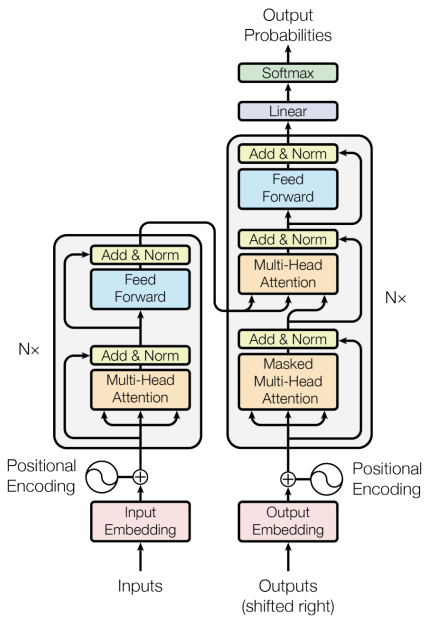

Figure 5: The Transformer architecture. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I., 2023. Attention is all you need. arXiv:1706.03762. Reprinted with permission.

In a Transformer, an operation called "self-attention" is used to integrate information from different parts of an input, which are often positions in a sequence (see Box 3 on attention; Vaswani et al. 2017). If we see the elements of the system which process information from each position (attention heads) as modules, then there is a basic similarity between the Transformer architecture and the global workspace: both integrate information from multiple modules. Transformers consist of a stack of layers of two types, which alternate: layers of attention heads, which perform the self-attention operation, moving information between positions, and feedforward layers. An interpretation of Transformers by Elhage et al. (2021) describes them as made up of "residual blocks", each consisting of one layer of each type, which process information drawn from and then added back to a "residual stream".

With the concept of the residual stream in mind, it is possible to argue that Transformers possess indicator properties GWT-1 through GWT-3—that is, that they have modules, a limited-capacity workspace introducing a bottleneck, and global broadcast. The residual stream

is the (alleged) workspace, and its dimensionality is lower than that of the self-attention and feedforward layers which write to it and read from it (Elhage et al. 2021). There is "global broadcast" in the sense that information in a particular layer in the residual stream can be used by downstream attention heads to influence further processing at any position. The information which is added to the residual stream at a given layer also depends on the state of the residual stream at earlier layers, so on this interpretation one might argue that Transformers meet the state-dependent attention requirement in GWT-4.

One problem with this argument is that it is questionable whether the dimensionality of the residual stream constitutes a bottleneck, since it is the same as that of the input to the system as a whole. However, a more fundamental problem with the argument is that Transformers are not recurrent. The residual stream is not a single network, but a series of layers interspersed between others. There are no modules that pass information to the residual stream and receive it back—instead, when attention heads and feedforward layers combine to write to the residual stream, this affects only what is received by different attention heads and feedforward layers downstream. One way to think about this is to ask which parts of Transformers are the modules. If modules are confined to particular layers, then there is no global broadcast. But if modules are not confined to particular layers, then there is no distinguishing the residual stream from the modules. Transformers lack the overall structure of a system with a global workspace, in that there is no one distinct workspace integrating other elements. There is only a relatively weak case that Transformer-based large language models possess any of the GWT-derived indicator properties.

The two versions of the Perceiver architecture, meanwhile, are closer to satisfying the GWT indicators than Transformers, but in our view still fall short of satisfying them all. The Perceiver architecture was designed to address a weakness of Transformers, which is that the self-attention operation integrates positions by generating pairwise interactions. This is computationally expensive to scale to high-dimensional inputs, motivating an approach that uses a single limited-capacity latent space to integrate information from specialists (Jaegle et al. 2021a, b, Goyal et al. 2022). In particular, Perceiver IO is designed to handle inputs from multiple domains or modalities, and to produce outputs of various kinds, using multiple input encoders and output decoders. It uses self-attention to process information in the latent space, and a related operation, cross-attention, to select information from input modules and write to output modules. The latent space alternates between self-attention and cross-attention layers, enabling it to take new information from input modules repeatedly. Perceiver IO achieves high performance in tasks including language processing, predicting movement in videos, image classification, and processing information about units for the purpose of action selection in Starcraft II (Jaegle et al. 2021b).

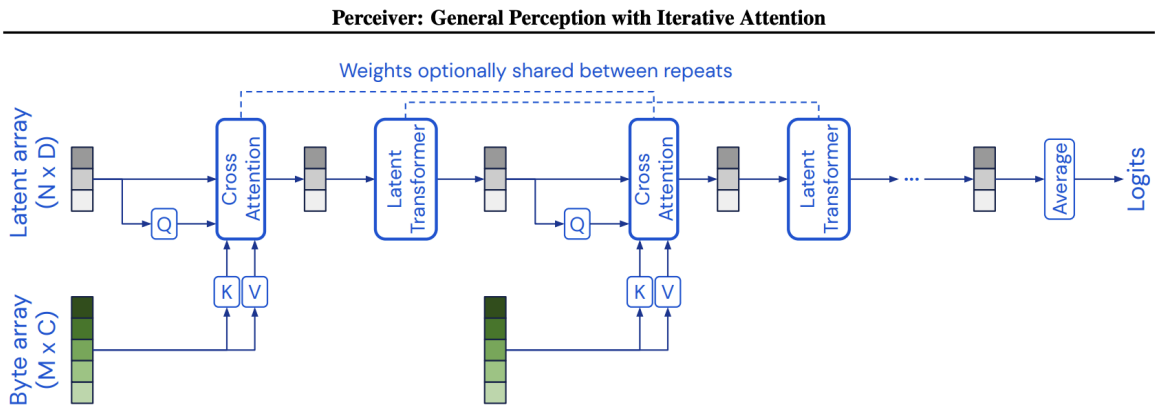

Figure 6: Perceiver architecture. Jaegle, A., Gimeno, F., Brock, A., Vinyals, O., Zisserman, A., & Carreira, J., 2021. In *Proceedings of the 38th International Conference on Machine Learning*, PMLR 139:4651-4664. Reprinted with permission.

The Perceiver architecture allows for sequences of inputs to be processed diachronically, with the latent space state updating with each new input, but also influenced by its previous state. So Perceiver arguably possesses indicator properties GWT-1 (specialised modules) and GWT-2 (bottleneck), as well as the first part of GWT-4 (state-dependent attention). However, it is notable that while specialised modules are possible in the architecture, they are not mandatory—the Perceiver can be used with one input network with a single function. Also, more importantly, the number of inputs that can be processed sequentially is limited by the number of cross-attention layers in the latent space. This means that although attention in Perceiver is state-dependent, in practice the system must be reset to begin a new task, so its states are determined by previous inputs on the current task.

As in the case of Transformers, the clearest missing element of the global workspace in the Perceiver is the lack of global broadcast. Perceiver IO has multiple output modules, but on any given trial, its inputs include an "output query", which specifies what kind of output is required. So only one output module acts on information from the workspace. Furthermore, input modules do not generally receive information from the workspace. So while the Perceiver architecture is important as an example of the successful use of a workspace-like method to improve functionality in AI, it is some way from being a full implementation of GWT.

3.2.2 Case studies for embodied agency

We now consider examples of AI systems that illustrate what is required for indicators AE-1 and AE-2, which concern agency and embodiment. The systems we discuss in this subsection are: PaLM-E (Driess et al. 2023), described as an "embodied multimodal language model"; a "virtual rodent" trained by RL (Merel et al. 2019); and AdA, a large Transformer-based, RL-trained "adaptive agent" (DeepMind Adaptive Agents Team 2023).